

Madhan Senthilkumar

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PROFESSIONAL SUMMARY

AI/ML enthusiast with 2 internships and hands-on experience across multiple ML projects spanning classification, regression, clustering, time-series forecasting, and deep learning. Deployed 2 live projects – a Random Forest-based customer churn prediction platform (81.5% accuracy) and an AI-assisted data cleaning tool (DataSanitizer Pro). Proficient in Python, Scikit-learn, TensorFlow, Pandas, and SQL, with practical experience across the ML lifecycle – EDA, feature engineering, model tuning, and evaluation. Comfortable working with AI development tools (Claude, ChatGPT) to accelerate coding and prototyping. IBM-certified in Python for Data Science.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning – Parul University, Vadodara

CGPA: 8.30 | Aug 2023 – 2027

EXPERIENCE

Data Science Intern

Future Interns | Dec 2025 – Jan 2026

- Designed a weighted scoring model across 5-point Likert scales to quantify student satisfaction, identifying a top-performing metric at 4.21/5 and flagging 3 critical areas below 3.6/5 for institutional review.
- Delivered an interactive visual report using Pandas and Seaborn translating raw survey responses into actionable recommendations for 2 key improvement areas.

Machine Learning Intern

Cognifyz Technologies (Remote) | Nov 2024 – Dec 2024

- Designed and executed end-to-end EDA and preprocessing pipelines using Python and Pandas on large-scale datasets, improving data reliability for downstream modeling.
- Built and evaluated multiple classification and regression models using Scikit-learn, selecting the best-performing model based on accuracy, precision, and F1-score metrics.

PROJECTS

Customer Churn Prediction & Retention Platform [Live Demo]

Python, Scikit-learn, SHAP, Streamlit | 2026

- Developed a hyperparameter-tuned Random Forest Classifier (81.5% accuracy, 0.854 ROC-AUC) predicting customer churn risk on 7,000+ Telco records.
- Integrated SHAP TreeExplainer for local feature attribution and built a vectorized batch-prediction engine for bulk CSV risk analysis.

DataSanitizer Pro [Live Demo]

Python, Pandas, NumPy, Next.js, TypeScript | 2025

- Built a data quality scoring engine that profiles missing values, duplicates, and statistical outliers (IQR/Z-score) across CSV/XLSX datasets, deployed live on Vercel.
- Automated per-column data quality reports with AI-generated fix suggestions; handled end-to-end ingestion, validation, and export workflows.

SafeGo – Inclusive Ride Assistance & Safety Platform [GitHub]

MERN, Firebase, ML, OSRM | 2025

- Embedded a Random Forest Classifier and Regressor for geospatial safety scoring, integrating ML predictions into a production backend via a Python inference layer.
- Designed real-time route navigation, distance, and ETA prediction pipelines using OpenStreetMap and OSRM data.

RetailAnalytics – Sales Intelligence Platform [GitHub]

Python, Scikit-learn, Prophet, SQL | 2026

- Built a K-Means customer segmentation model on RFM-style transaction data to cluster customers into 3 behavioral segments.
- Implemented Isolation Forest anomaly detection and Facebook Prophet time-series forecasting, evaluated via MAE, RMSE, and R-squared.

TECHNICAL SKILLS

Machine Learning: Classification, Regression, Clustering, Random Forest, K-Means, Isolation Forest, Model Evaluation, Hyperparameter Tuning, Feature Engineering

Data & Analysis: Pandas, NumPy, EDA, Scikit-learn, Prophet, SQL, MongoDB, MySQL

Programming: Python, Java, JavaScript, TypeScript, SQL

AI Tools: Claude, ChatGPT (AI-assisted coding & prototyping)

Tools & Platforms: Git, GitHub, Docker, Jupyter Notebook, Vercel, Streamlit

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Developer – Associate (Infosys Springboard)
- Oracle – Agentic AI Certified Foundations Associate
- Anthropic – Claude Code 101
- HackerRank – SQL (Advanced) Certificate
- IBM – Python for Data Science (2025)